

VizR Workbench User Manual

The workbench is the basic working unit of RNA-Seq analysis. This manual explains the workbench creation process and the function of each screen.

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Part 1. Workbench Creation Guide

Workbench creation consists of three steps: **Basic Setup** → **Sample Mapping** → **Pipeline Configuration**

Step 1. Basic Setup

Click the **Create New Workbench** button on the dashboard to open the creation modal.

1-1. Workbench Name

Enter the name of the analysis project.

- Allowed characters: **English letters (A-Z, a-z), numbers (0-9), underscore (_), hyphen (-)**
- Korean characters, spaces, and special characters are removed automatically
- Duplicate names are validated in real time while typing (green check = available)

Example: Cold_Stress_Experiment, drought-response-2024

 **Warning:** The workbench name is also used as a directory name on the server. A concise name that indicates the analysis purpose is recommended.

1-2. Raw Data Source

Select one of the three tabs for the data input method. Only the data from the **selected tab** is used in the workbench.

 **Warning:** Even if data is entered in multiple tabs, only the data from the currently active tab is used. If another tab already contains data, a notice message appears at the bottom.

(A) Local Upload

Upload FASTQ/FASTA files directly from the local PC.

Supported formats: `.fastq` , `.fq` , `.fasta` , `.fa` , and compressed `.gz` files

How to use:

1. Drag and drop files into the upload area, or click to select them
2. Upload progress is displayed, and a green check appears after completion
3. All uploads must be completed before moving to the next step

 **Warning:** RNA-seq files are often larger than 1GB. Do not close the browser during upload. The Next button remains disabled until the upload is complete.

 **Warning:** If the modal is closed, all uploaded files are deleted. Be careful not to close it by mistake.

(B) NCBI BioProject

Search and import data from NCBI SRA using a BioProject ID.

How to use:

1. Enter a BioProject ID (example: `PRJNA252931`)
2. Click the **Search** button
3. Select the files to use from the result table using checkboxes
4. The layout (SE/PE) is detected and configured automatically

File selection options:

- Top checkbox: select or deselect all
- For mixed SE/PE BioProjects, only SE or PE files can be selected in bulk using the corresponding buttons

⚠ **Warning:** SE (Single-End) and PE (Paired-End) files cannot be selected together. Select files with the same layout only.

⚠ **Warning:** Long-read data such as PacBio or Nanopore is detected automatically and excluded. VizR supports Illumina short-read data only.

(C) From Server

Specify FASTQ/FASTA files that already exist on the server by path.

⚠ **Warning:** Files recognized by VizR are restricted to the **VizR Data Folder** (`/vizr`). Files in other server locations cannot be accessed by VizR, so they must first be copied or moved into the VizR Data Folder before the path is entered.

How to enter file paths:

Files can be recognized by VizR when they are placed inside the **VizR Data Folder** selected during VizR installation.

i Reference: The VizR Data Folder is the folder selected during the "Select VizR Data Folder" step at installation. The default on Windows is `C:\Users\username\Documents\VizR_Data` , and the default on Linux is `~/VizR_Data` .

Inside VizR, this folder is displayed as the `/vizr` path. Therefore, file paths must begin with `/vizr/` .

Actual location on your computer	Path entered in VizR
<code>VizR_Data\raw_data\sample1.fastq.gz</code>	<code>/vizr/raw_data/sample1.fastq.gz</code>
<code>VizR_Data\experiment\sample2.fq.gz</code>	<code>/vizr/experiment/sample2.fq.gz</code>

How to use:

1. Copy the FASTQ files to analyze into the **VizR Data Folder** (subfolders can be created)

2. Enter one `/vizr/` path **per line** in the text area
3. When Next is clicked, file existence and format are validated automatically

Example: if a `raw_data` folder was created inside the VizR Data Folder and files were placed there

```
/vizr/raw_data/sample1_R1.fastq.gz  
/vizr/raw_data/sample1_R2.fastq.gz  
/vizr/raw_data/sample2_R1.fastq.gz  
/vizr/raw_data/sample2_R2.fastq.gz
```

 **Warning:** Files outside the VizR Data Folder cannot be accessed by VizR. Copy the files into that folder first, then enter them as `/vizr/...` paths.

 **Warning:** If long-read data is included, it is blocked during validation.

1-3. Analysis Species

Select the target species for analysis.

- Currently, only **Arabidopsis thaliana** is enabled
- Lemna, Spirodela, Wolffia, and others are planned for future support

1-4. Move to the Next Step

After entering all items, click the **Next** button to move to Step 2 (Sample Mapping).

The following items are validated automatically when Next is clicked:

- Workbench Name is empty or duplicated
- No file exists in the selected data source
- A Server path file does not exist or has an invalid format

If validation fails, a red error message appears at the top. Correct the relevant item and try again.

Step 2. Sample Mapping

This step maps the uploaded FASTQ files to experimental groups and samples.

2-1. Data Layout

Select the sequencing layout of the data.

Layout	Description	Number of files
Single-End (SE)	One FASTQ file per sample	1 file/sample
Paired-End (PE)	Two files per sample: Forward + Reverse	2 files/sample

- For NCBI data, the automatically detected value from Step 1 is already set
- For Local Upload / From Server, it must be selected manually

 **Warning:** If the layout is changed, all existing file assignments are **reset**. A confirmation popup appears, so change it carefully.

2-2. Available Files (File Assignment)

The **Available Files** panel shows the list of files that can be mapped. Assign them to the mapping table below using drag and drop.

- Drag a file and drop it onto the target row assignment area to complete the assignment
- Click the **X** button next to an assigned file to remove it and return it to Available Files
- When all files are assigned, the message "All files have been assigned" is displayed

 **Warning:** When the NCBI data source is selected, files are already assigned automatically in the mapping table, so this panel is empty. Only the Group Name needs to be entered.

2-3. Sample Mapping Table

Each row represents one sample. The number of rows is calculated automatically according to the number of files and the selected layout.

Column	Input method	Description
Group Name	Manual input	Experimental group name (example: <code>Control</code> , <code>Treatment</code> , <code>Cold_6h</code>)
Replicate	Manual input	Sample name (example: <code>Rep1</code> , <code>Rep2</code>). Must be unique across the workbench
FASTQ File (SE)	Drag and drop	FASTQ file for that sample
Forward Read (PE)	Drag and drop	Forward read file for Paired-End (<code>_1.fastq.gz</code>)
Reverse Read (PE)	Drag and drop	Reverse read file for Paired-End (<code>_2.fastq.gz</code>)

- **Add Row:** add a mapping row using the button at the upper right
- **Remove Row:** delete a row using the trash icon on the right side of each row

 **Warning:** Group Name is used as the comparison group in DEG analysis. Samples from the same condition must use exactly the same Group Name.

2-4. Move to the Next Step

The following items are validated when **Next** is clicked:

- Unassigned files remain in Available Files
- Sample Name (Replicate) is empty or duplicated
- Group Name is empty

After validation passes, the workflow moves to Step 3 (Pipeline Configuration).

Step 3. Pipeline Configuration

Configure the tools and parameters for each stage of the RNA-Seq analysis pipeline.

3-1. Reference Genome Settings

Configure the reference genome used for Alignment and Quantification.

Item	Description
Reference Set	Select the reference genome version (example: <code>TAIR10</code>). The list is filtered according to the species selected in Step 1
Species	The species selected in Step 1 is displayed automatically (read-only)

 **Warning:** If no Reference Set is available for the selected species, it must be uploaded first on the Settings page.

3-2. Quality Control

Evaluate the quality of the sequencing data.

Tool	Description	Parameters
FastQC	Generate a quality report for Illumina sequencing data	<code>threads</code> : number of processing threads (default: 4)
Skip QC	Skip the QC stage	None

3-3. Sequence Cleaning

Remove low-quality reads and perform adapter trimming. Choose one of four options.

(A) Skip Cleaning

Use the raw data without preprocessing.

(B) Trimmomatic Only

Perform Illumina adapter removal and quality-based trimming.

Parameter	Default	Description
<code>leading</code>	3	Minimum quality at the 5' end

Parameter	Default	Description
<code>trailing</code>	3	Minimum quality at the 3' end
<code>slidingwindow</code>	4:15	Sliding window (window size:min quality)
<code>minlen</code>	36	Minimum read length

(C) PRINSEQ Only

Perform quality filtering and trimming.

Parameter	Default	Description
<code>min_len</code>	50	Minimum sequence length
<code>min_qual_mean</code>	20	Minimum mean quality score
<code>trim_qual_left</code>	20	Quality-based trimming at the 5' end
<code>trim_qual_right</code>	20	Quality-based trimming at the 3' end

(D) Trimmomatic → PRINSEQ (Sequential)

Apply PRINSEQ sequentially after Trimmomatic. Parameters for both tools are configured separately.

When this option is selected, additional **ILLUMINACLIP** settings are displayed:

Item	Default	Description
Adapter	TruSeq v3	Adapter type (TruSeq v3 / TruSeq v2 / Nextera / None)
seed	2	Allowed number of seed mismatches
palindrome	30	Palindrome clip threshold
simple	10	Simple clip threshold

 **Warning:** Nextera adapters are for Paired-End data only.

3-4. Quantification

Perform read alignment and gene expression quantification.

Tool combination	Description	Main parameters
HISAT2 + StringTie	Splice-aware alignment + transcript-assembly-based quantification	<code>threads</code> (8), <code>max_intronlen</code> (500000), <code>min_coverage</code> (2.5), <code>min_transcript_len</code> (200)
Bowtie + RSEM	Fast alignment + integrated RSEM quantification	<code>threads</code> (4), <code>max_mismatches</code> (2)
Bowtie2 + RSEM	Sensitive alignment + integrated RSEM quantification	<code>threads</code> (4), <code>preset</code> (sensitive / very-fast / fast / very-sensitive)

i Reference: If Bowtie/Bowtie2 + RSEM is selected, alignment and quantification are handled together, so the separate Count stage is disabled automatically.

3-5. Differential Expression

Perform differential expression gene (DEG) analysis.

Tool	Description	Parameters
edgeR	Empirical analysis of digital gene expression data	<code>fdr</code> : False Discovery Rate (default: 0.05), <code>logfc</code> : Log Fold Change threshold (default: 1)

3-6. Create Workbench

After confirming all settings, click the **Create Workbench** button to create the workbench.

- When creation completes, the modal closes automatically and the page moves to the corresponding workbench
- During creation, the button changes to the "Creating..." state and duplicate clicks are blocked
- If creation fails, an error message is displayed. Review the message, correct the issue, and try again

i Reference: Creating a workbench only saves the configuration. The actual analysis pipeline is started separately from the workbench detail page using the **Start Analysis** button.

Part 2. Screen Guide

1. Workbench Overview (Workbench List)

This is the main screen shown after login, where the list of created workbenches can be viewed.

Screen Layout

Area	Description
Workbench Table	Displays existing workbenches with Name, Status, Created At, and Last Updated columns. Clicking a row moves to the detail page
Empty-state guide	Displays a creation guide message when no workbench exists

Buttons

Button	Description
Create Workbench	Upper-right corner. Opens the workbench creation modal (→ Step 1)
Create Your First Workbench	Displayed at the center of the screen when no workbench exists. Same function

2. Overview (Workbench Detail)

This detail screen is shown when a workbench is clicked. It provides a consolidated view of workbench information, pipeline status, data summary, and pipeline configuration.

Header Area

Displays the basic information of the workbench.

Item	Description
Workbench Name	Name entered at creation time

Item	Description
Species	Target species of the analysis
Created / Updated	Creation time and last update time
Created by	Name of the creator

Header buttons:

Button	Description
Share	Share the workbench with another user. Shared users can view results in read-only mode
Edit Pipeline	Modify pipeline settings. Disabled while the pipeline is running
Delete	Delete the workbench

i Reference: For a shared user, the Share button is displayed as **Leave**, and the Edit Pipeline and Delete buttons are disabled.

Pipeline Status Area

Displays the current pipeline execution status.

Status	Description
Not Started	Analysis has not been started yet
Pipeline Starting	Pipeline startup is being prepared
Pipeline Running	Analysis is in progress. A progress bar and current step are displayed
Pipeline Completed	All analysis stages are complete
Pipeline Failed	Analysis failed. An error message is displayed

- Progress bar: number of completed steps / total steps (example: **1/4 steps**)
- Start time and completion time are displayed

Pipeline buttons:

Button	Description
Start Analysis	Start pipeline execution. Disabled if already running
Stop Analysis	Stop the running pipeline. Enabled only while running

Data Overview Area

Displays summary information about workbench data as cards.

Card	Description
Data Layout	Sequencing layout (Single-End or Paired-End)
Samples	Number of mapped samples
Files	Number of raw data files

Pipeline Steps Area

Displays configured pipeline stages in sequence as cards. Each card shows the stage number, description, and tool used.

Example: ① Data Download (data_downloader) → ② Alignment (hisat2) → ③ Quantification (stringtie) → ④ DEG (edger)

i Reference: The pipeline step composition reflects the settings defined in Step 3 when the workbench was created. If changes are needed, use the **Edit Pipeline** button.

3. Raw Data (Raw Data)

This screen is used to check download/copy progress during pipeline execution, data source information, and sample mapping information.

Real-time File Progress Area

Monitors file download (or copy) progress in real time. Updates are delivered automatically through WebSocket.

Summary cards:

Card	Description
Total Files	Total number of files

Card	Description
Downloading	Number of files currently being downloaded
Completed	Number of files whose download is complete
Connection	WebSocket connection status (Live / Connecting / Offline)

Per-file progress:

Below the summary cards, the progress status of each file is displayed as a card.

Item	Description
File name	Name of the target file
Status	File status (<code>pending</code> / <code>downloading</code> / <code>completed</code> / <code>compressed</code> / <code>failed</code>)
Progress	Percentage and downloaded size / total size
Progress bar	Shows a spinner before the download starts, then switches to a progress bar after start

i Reference: If the pipeline has not started yet or the current stage is not the download stage, the message "Loading file information..." is displayed.

Data Source Information Area

Displays the data source information configured during workbench creation.

Item	Display condition	Description
Input Method	Always	Data input method (<code>local upload</code> / <code>ncbi</code> / <code>server</code>)
NCBI BioProject	When the source is NCBI	BioProject ID (example: <code>PRJNA252931</code>)
Server Paths	When the source is Server	Number of configured server paths
Layout Type	Always	Sequencing layout (Paired-End or Single-End)

Sample Mapping Information Area

Displays the sample mapping information configured in Step 2 during workbench creation as a table.

Column	Description
Group	Experimental group name
Sample	Sample name
Forward Read	Forward read file name
Reverse Read	Reverse read file name (displayed only for Paired-End)

i Reference: This table is read-only. To modify sample mapping, a new workbench must be created.

4. Quality Control (Quality Control)

This screen is used to review FastQC-based sequencing data quality analysis results. The screen layout changes according to the pipeline status.

i Reference: If the QC stage was configured as "Skip QC" during workbench creation, this tab shows a disabled-state message.

Screen by Analysis Status

Status	Screen content
Not Started / Pending	Message indicating that data download completion is pending
Running	Analysis progress display (number of completed files / total files, progress bar)
Completing	Completion message (automatically switches to the result screen after about 3.5 seconds)
Completed	QC result dashboard (described in detail below)

Completed State - Result Dashboard

After analysis is completed, the screen switches to a left sidebar + right main area layout.

Left sidebar:

Item	Description
Overview	Move to the overall sample summary view
Sample list	Each sample is displayed with a quality-status badge (green=Passed, yellow=Warning, red=Failed). Clicking an item moves to the sample detail view

Each sample item also displays Total Sequences and Avg Quality.

Right main area - when Overview is selected:

Area	Description
Module Statistics	Summary cards for Total Samples, Total Modules, and Passed/Warning/Failed Modules
Sample Cards	Displays all samples in a card grid. Each card contains sample name, status (PASS/WARN/FAIL), group, Total Sequences, Avg Length, and Avg Quality. Clicking a card moves to the detail view

Right main area - when an individual sample is selected:

This view is composed of three layers.

Layer 1. Core metrics summary:

- **Summary Statistics:** Total Sequences, Average Length, Average Quality, Sample ID
- **Module Statistics:** Table of Pass/Warn/Fail state for each FastQC module
- **Quick Summary:** Summary of AI analysis results (when enabled in Settings)

Layer 2. AI quality analysis (Detailed Quality Analysis):

- Displays AI-generated quality evaluation, concerns, and recommendations in markdown format
- If the AI analysis feature is disabled in Settings, a guidance message is displayed instead

Layer 3. Quality Metrics charts:

Visualization charts for each FastQC module. The module status (Pass/Warn/Fail) is shown at the top of each chart.

Chart	Description
Per Base Sequence Quality	Distribution of quality scores by read position
Per Sequence Quality Scores	Distribution of average quality scores by sequence
Per Base Sequence Content	A/T/G/C ratio by read position
Per Sequence GC Content	GC content distribution by sequence

Chart	Description
Per Base N Content	N (undetermined base) ratio by read position
Sequence Length Distribution	Sequence length distribution
Adapter Content	Adapter sequence contamination ratio
Sequence Duplication Levels	Sequence duplication level

i Reference: Charts for modules without data are hidden automatically.

5. Preprocessing (Preprocessing)

This screen is used to review sequence preprocessing results generated by Trimmomatic and PRINSEQ. Use the sub-tabs at the top to switch between the two tools.

i Reference: If Sequence Cleaning was set to "Skip Cleaning" during workbench creation, the corresponding tool tab displays a disabled-state message. Only the configured tool is enabled.

Sub-tabs

Tab	Description
Trimmomatic	Trimmomatic preprocessing progress and results
PRINSEQ	PRINSEQ preprocessing progress and results

A status indicator is shown next to each tab (gray=Pending, blinking green=Running, green=Completed, red=Failed). While the pipeline is running, the screen automatically switches to the tab of the currently running tool.

Trimmomatic Tab

Common area:

- Header: tool name, WebSocket connection status (Live/Offline), analysis status badge
- Progress bar: number of completed files / total files, percentage

Result area after completion:

Summary statistic cards:

Card	Description
Total Samples	Number of processed samples
Input Reads	Total input reads
Surviving Reads	Number of reads that passed quality filtering
Survival Rate	Average survival rate (%)

Per-sample result cards: each card shows Input Reads, Surviving, and Removed counts with a survival-rate bar. The color changes according to the survival rate (>=90%=green, >=80%=yellow, <80%=red).

PRINSEQ Tab

Uses the same layout as Trimmomatic.

Result area after completion:

Summary statistic cards:

Card	Description
Total Samples	Number of processed samples
Input Sequences	Total input sequences
Good Sequences	Number of sequences that passed quality filtering
Quality Rate	Average quality pass rate (%)

Per-sample result cards: each card shows Input Sequences, Good Sequences, and Bad Sequences. Length Filtered and Quality Filtered counts are also displayed when available. The color changes according to the quality pass rate (>=95%=green, >=90%=yellow, <90%=red).

6. Alignment (Alignment)

This screen is used to review read alignment results against the reference genome.

Screen Layout**Header area:**

Item	Description
Read Alignment	Screen title
Tool badge	Displays the name of the alignment tool used (example: HISAT2)
WebSocket status	Real-time connection status (Live/Offline)
Status badge	Analysis status (Not Started / Pending / Running / Completed / Failed)

Progress bar: displays the number of completed files / total files and the percentage

Result Area After Completion

Summary statistic cards:

Card	Description
Overall Mapping Rate	Overall alignment rate (%)
Total Reads	Total input reads (in millions)
Mapped Reads	Number of reads aligned to the reference (in millions)
Unmapped Reads	Number of reads not aligned (in millions)

Per-sample result cards: each card shows Total Reads, Mapped Reads, and Unmapped Reads with a mapping-rate bar. If Multi-mapped Reads exist, they are shown additionally. The color changes according to the mapping rate (>=90%=green, >=80%=yellow, <80%=red).

i Reference: Samples with mapping rates below 80% should be reviewed for data quality or reference suitability.

7. Counts (Expression Data)

This is an interactive screen for viewing, searching, and analyzing gene expression quantification results.

i Reference: Data appears on this screen only after quantification is complete. Before completion, a progress bar and waiting message are displayed.

1. Matrix Type Selection

Use the button group at the upper right to switch the expression data type.

Type	Description
Gene Count (Raw)	Raw read count at the gene level
TPM	Transcripts Per Million normalization
TMM	Trimmed Mean of M-values normalization (default)

- The table is refreshed automatically when the type changes
- Search terms and pattern-analysis state are reset when the type changes

2. Gene Search

Genes can be searched in the search area at the upper left.

How to use:

1. Enter a Gene ID or Gene Symbol in the search box
2. To search multiple genes at once, separate them with **commas, spaces, tabs, or line breaks**
3. Click the **Search** button or press **Ctrl+Enter** to execute the search
4. Click the **Clear** button to reset the search and return to all data

Example: AT1G01010, AT1G01020, AT1G01030

When the search is active, a blue information bar appears above the table and shows the number of matched genes.

3. Pattern-Based Gene Selection

This function automatically searches for genes that follow a specific expression pattern. It is implemented as a collapsible/expandable panel.

Audio equalizer UI:

For each experimental group (condition), set the expected relative expression level using sliders (scale 0 to 10). For example, if Control=2, Treatment_6h=5, and Treatment_24h=8, genes that increase over time are searched.

Presets:

Preset	Description
Weak Up	Weak up-regulation pattern (3→6)
Strong Up	Strong up-regulation pattern (2→9)
Weak Down	Weak down-regulation pattern (6→3)
Strong Down	Strong down-regulation pattern (9→2)
Custom	Set directly by the user

Each group can be disabled individually (OFF), and disabled groups are excluded from the analysis.

Statistical Thresholds:

Parameter	Default	Description
Min Spearman Correlation	0.6	Minimum correlation with the specified pattern (higher is stricter)
Min Log2 Fold Change	1.0	Minimum expression change between groups
Min CPM	0	Minimum expression level in at least one group (Counts Per Million)

Constraint Tolerances:

Parameter	Default	Description
Equality Tolerance	0.3	Allowed difference between groups set to the same level (log2 scale)
Order Gap	0.5	Minimum difference between groups with different ordering (log2 scale)

CV Filter (optional, enabled by toggle):

Parameter	Default	Description
Max CV	0.5	Maximum coefficient of variation within a group (SD/mean)
Min Groups Pass CV	1	Minimum number of groups that must pass the CV threshold

Equality Constraints (optional):

Specify pairs of groups that should have similar expression levels. Select two groups and add them with the **Add** button.

Execution:

- The **Analyze Pattern** button is enabled only when at least two groups are active
- Matching genes are displayed in the table
- If no result is found, a guide modal recommending threshold adjustment is displayed
- The **Reset All** button resets all settings

 **Warning:** If a search is performed while pattern analysis is active, the search is applied only within the pattern result. To return to the full dataset, click the **Reset to all genes** link in the information bar.

4. Expression Table

Displays gene expression data in table form.

Table columns:

Column	Description	Interaction
Checkbox	Select/deselect genes	Use individual clicks or the header select-all checkbox
Gene ID	Gene identifier (example: AT1G01010)	Hover to show the Gene Description tooltip
Gene Symbol	Gene symbol (example: NHX1)	Displayed in italics only when it differs from the Gene ID
Expression Pattern	Mini heatmap visualization	Shows the expression pattern of the gene across all samples as colors
Sample Columns	Expression value for each sample	Clicking a column header highlights that column (blue background)

- The Gene ID and Gene Symbol columns are fixed on the left (Sticky), so they remain visible during horizontal scrolling
- Table height adjusts automatically to the browser viewport

5. Pagination

Page navigation is displayed at the top of the table.

Item	Description
Rows per page	Select rows per page (100 / 500 / 1000 / 2000 / 5000)

Item	Description
Displayed range	Current visible row range (example: 1-500 of 27,416)
Page number	Click to move to that page. Collapsed display is used when more than 7 pages exist
Previous / Next	Move to the previous/next page

6. Analyze Selected Genes (Analyze Menu)

After selecting genes with checkboxes in the table, click the **Analyze (N)** dropdown button to run the following analyses.

Analysis	Description
Heatmap	Visualize expression of the selected genes as a heatmap
GO Enrichment	Gene Ontology functional analysis (BP/MF/CC)
KEGG Pathway	KEGG pathway analysis
Venn Diagram	Venn diagram comparison

i Reference: The Analyze button is enabled only when at least one gene is selected. The number in parentheses indicates the number of currently selected genes.

7. Download

Click the **Download** button (green) to download the current expression data for the active matrix type as a file.

- If a search is active, only the filtered results are downloaded (**Filtered** label on the button)
- If no search is active, the entire dataset is downloaded

8. DEG (Differential Expression)

This screen is used to explore differential expression gene (DEG) analysis results. Results produced by DESeq2 or edgeR can be reviewed through multiple visualizations and tables.

1. Analysis Status

When entering the DEG screen, different views are displayed depending on the pipeline state.

Status	Screen
not_started	"No DEG Results Available" - guidance to run DEG analysis in the pipeline first
pending / running	"Analysis in Progress" with a spinner. The analysis tool name (DESeq2/edgeR) is shown. Status refreshes automatically every 5 seconds
failed	"Analysis Failed" - guidance to inspect pipeline logs or rerun
insufficient_replicates	DESeq2-specific warning. At least 2 biological replicates per group are required. The current sample structure (group name and sample count) is shown, and edgeR use or additional replicates are recommended
completed	Analysis result dashboard is displayed (described in detail below)

 **Warning:** The `insufficient_replicates` state occurs only in DESeq2. If a group has only one replicate, analysis may still be possible by using edgeR.

2. Screen Layout (completed state)

After the analysis is completed, the screen is displayed with the following layout.

Header area:

- "Differential Expression Analysis" title
- Analysis tool badge (DESEQ2 or EDGER, purple badge)

Left sidebar: Comparisons

A list of comparison conditions is displayed. Buttons are listed for each group combination configured in the workbench (example: Treatment_vs_Control).

- The selected comparison is highlighted with a blue background

- Each button shows the comparison name and the analysis tool name
- Changing the comparison resets the selected genes automatically
- The sidebar is hidden in the Expression Matrix tab (because it is not comparison-specific)

Right content: tab navigation

Five tabs are provided:

Tab	Description
Matrix (Up)	Table of upregulated genes ($\logFC > 0$, $FDR < 0.05$)
Matrix (Down)	Table of downregulated genes ($\logFC < 0$, $FDR < 0.05$)
MA Plot	Scatter plot of mean expression ($\logCPM$) vs change ($\logFC$)
Volcano Plot	Scatter plot of change ($\logFC$) vs statistical significance ($-\log_{10} p\text{-value}$)
Expression Matrix	Expression matrix filtered by P-value / Fold-change criteria

i Reference: Selected genes are reset when switching tabs. Gene selections are not preserved across tabs.

3. Matrix (Up) / Matrix (Down) Tabs

Display only upregulated or downregulated genes in a table. Both tabs use the same structure and functions; only the filtering criterion differs.

Toolbar area (above the table):

Element	Description
Search	Search Gene ID or Gene Symbol. Multiple genes can be entered in a textarea separated by commas/spaces/tabs/line breaks. Press Ctrl+Enter to search
Search button	Execute search
Clear button	Displayed only when search text exists. Resets the search
Analyze button	Analysis dropdown for selected genes (shows the number selected in parentheses). Disabled when no gene is selected
Download button	Download in TSV format. If genes are selected, only those genes are downloaded; otherwise the current page is downloaded
Pagination	Rows per page (100/500/1000/2000/5000), page number, Previous/Next

Analyze dropdown menu (Matrix Up/Down tabs):

Analysis	Description
Heatmap	Expression heatmap of the selected genes
GO Enrichment	Gene Ontology functional analysis (BP/MF/CC)
KEGG Pathway	KEGG pathway analysis
Venn Diagram	Venn diagram comparison of the selected gene set

Data table:

Column	Description
Checkbox	Gene selection (individual/select all). Sticky column
Gene ID	Gene ID. Hover to show the Gene Description tooltip. Sticky column
Gene Symbol	Gene symbol (italic). Shown as plain text when identical to the Gene ID. Sticky column
Expression Pattern	MiniHeatmap based on TMM-normalized values. Visualizes per-sample expression pattern as color bars
logFC	Log2 Fold Change. Filterable/sortable (click the filter icon in the column header)
logCPM	Log2 Counts Per Million. Filterable/sortable
PValue	Raw P-value (scientific notation). Filterable/sortable
FDR	False Discovery Rate (scientific notation). Filterable/sortable

Column filter & sorting function:

Click the slider icon at the right of the **logFC** , **logCPM** , **PValue** , or **FDR** column header to open the **Filter & Sort popup**.

Sorting options:

Option	Description
Ascending	Ascending order (small → large)
Descending	Descending order (large → small)

Value filter options (Filter by Value):

Operator	Meaning	Example
Greater than (>)	Greater than the entered value	logFC > 2

Operator	Meaning	Example
Greater than or equal ($\geq$)	Greater than or equal to the entered value	<code>logFC $\geq$ 1</code>
Less than ($<$)	Less than the entered value	<code>FDR $<$ 0.01</code>
Less than or equal ($\leq$)	Less than or equal to the entered value	<code>PValue $\leq$ 0.05</code>
Equal to ($=$)	Equal to the entered value	<code>logCPM = 0</code>

When an operator is selected, a numeric input field appears and accepts decimal values. Use the **Apply** button to apply the filter, and the **Clear** button to reset both filtering and sorting for that column.

Active-state indicators:

- While a filter is active: a blue dot appears at the upper-right of the icon (pulse animation)
- While sorting is active: an $\uparrow$ / $\downarrow$ arrow appears next to the column name, and the icon changes to blue

i Reference: Filtering and sorting are server-side operations. Applying a filter triggers a new query against the full dataset.

i Reference: Filters can be applied to multiple columns simultaneously, and the conditions are combined with **AND**. Example: if `logFC > 2` and `FDR < 0.01` are both set, only genes satisfying both conditions are displayed.

Download file format:

- Format: TSV (Tab-Separated Values)
- Example file name: `Treatment_vs_Control_up_regulated_selected_25_2026-02-13.tsv`
- Included columns: `GeneID, GeneSymbol, logFC, logCPM, PValue, FDR`

4. Volcano Plot Tab

Visualizes `logFC` (x-axis) vs `-log10(p-value)` (y-axis) as a scatter plot.

Chart area:

Element	Description
Title	<code>"Volcano Plot: {comparison name}"</code>

Element	Description
Chart display limit	Dropdown in the upper right to select the number of significant genes displayed in the chart: Top 100 / 300 / 500 / 1,000 / 2,000 / All
Statistics summary	Displays Total genes, Upregulated (red), and Downregulated (blue) counts
X-axis	Log Fold Change (logFC)
Y-axis	-log10(p-value)
Threshold lines	Two vertical dotted lines: logFC = 1 (FC > 2), logFC = -1 (FC < 0.5). One horizontal dotted line: p-value = 0.05

Point color categories:

Color	Condition	Size
Red	Significant + logFC > 1 (upregulated)	Large points
Blue	Significant + logFC < -1 (downregulated)	Large points
Orange	Significant but `	logFC
Gray	Non-significant	Small points (semi-transparent)

Tooltip (when hovering over a point):

- Gene Symbol (italic), Gene ID
- logFC , -log10(p-value) , and FDR values
- Significance state (Upregulated / Downregulated / Significant / Not significant)
- For significant genes, an Expression Pattern MiniHeatmap is displayed additionally

Interpretation panel (below the chart):

- Text guide for interpreting the chart
- When the chart display limit is smaller than the number of all significant genes, a note is shown (example: "Chart displays top 100 significant genes. All 1,234 significant genes are available in the table below.")
- Summary counts for Significant / Up / Down

Significant Genes Table (below the chart):

The table in the Volcano Plot tab displays only significant genes with FDR < 0.05 .

Column	Description
Checkbox	Gene selection
Gene ID	Gene ID (Description tooltip on hover)
Gene Symbol	Gene symbol (italic)
Expression Pattern	TMM-based MiniHeatmap
logFC	Filterable/sortable (red: positive, blue: negative)
logCPM	Filterable/sortable
PValue	Filterable/sortable (scientific notation)
FDR	Filterable/sortable (scientific notation)
Regulation	Upregulated (red) / Downregulated (blue) / Significant (orange)

Search, Analyze, Download, and Pagination work the same as in the Matrix tabs. However, the Analyze menu does not include Venn Diagram.

5. MA Plot Tab

Visualizes mean expression (**logCPM** , x-axis) vs change (**logFC** , y-axis) as a scatter plot.

Chart area:

Element	Description
Title	"MA Plot: {comparison name}"
Chart display limit	Same as Volcano Plot (Top 100 ~ All)
Statistics summary	Displays Total genes and Significant (red) counts
X-axis	Average Expression (logCPM)
Y-axis	Log Fold Change (logFC)

Point color categories:

Color	Condition
Red	Significant gene (FDR < 0.05). Large point, opaque
Gray	Non-significant gene. Small point, semi-transparent

Tooltip (when hovering over a point):

- Gene ID, Gene Symbol

- `logCPM` , `logFC` , and `FDR` values
- Significance state (`Significant / Not significant`)
- For significant genes, an Expression Pattern MiniHeatmap is displayed additionally

Interpretation panel: same structure as the Volcano Plot. Includes a chart interpretation guide and statistics summary.

Significant Genes Table: same structure as the Volcano Plot tab. In the Regulation column, `logFC > 0` is classified as Upregulated and `logFC < 0` as Downregulated.

6. Expression Matrix Tab

This tab allows the user to generate an expression matrix of differential genes by setting P-value and Fold-change thresholds directly.

i Reference: The left Comparisons sidebar is hidden in this tab. Expression Matrix integrates the overall DEG results of the workbench instead of a single comparison condition.

Parameter input area (top panel):

Parameter	Description	Default
P-value Cutoff	Significance threshold. Range <code>0~1</code> , step <code>0.01</code>	<code>0.01</code>
Fold-change Cutoff (log2)	Change threshold. Integer step	<code>2</code>
Generate Matrix button	Generate an expression matrix using the configured parameters. Internally runs <code>analyze_diff_expr.pl</code>	

- After generation completes, the current matrix information is displayed in the form `"Current Matrix: P=0.01, C=2"`
- If newly generated, a green `"Newly generated"` check label is shown
- After changing parameters and clicking Generate Matrix, search, selection, and page state are all reset

Data table:

Column	Description
Checkbox	Gene selection. Sticky column
Gene ID	Gene ID. Sticky column. Description tooltip on hover

Column	Description
Gene Symbol	Gene symbol (italic). Sticky column
Expression Pattern	Visualizes per-sample expression values as a MiniHeatmap
Sample columns	<code>log2 centered</code> expression values for each sample (2 decimal places)

- Rows per page: `100 / 500 / 1000` (fewer options than the Matrix tabs)
- Individual `logFC / PValue / FDR` filter functions are not provided (the entire dataset is filtered using the P-value/Fold-change cutoff instead)

Download:

- Clicking the Download button triggers direct CSV download from the server (POST request)
- If genes are selected, only those genes are downloaded; otherwise the full matrix is downloaded
- Example file name: `diffExpr.P001_C2.selected_50_genes.csv` or `diffExpr.P001_C2.matrix.log2.centered.csv`

Analyze dropdown menu (Expression Matrix tab):

Analysis	Description
Heatmap	Expression heatmap of the selected genes
GO Enrichment	Gene Ontology functional analysis
KEGG Pathway	KEGG pathway analysis
Venn Diagram	Venn diagram comparison

7. Common Interaction Pattern

Gene selection:

- Individual selection: click the checkbox of each row
- Select all: use the header checkbox to select/deselect all genes on the current page
- Selection state is reset when the page changes, the tab changes, or the comparison changes
- The number of selected genes is reflected in real time inside the Analyze button parentheses

MiniHeatmap:

- Small heatmap shown in each gene row
- Color representation based on TMM-normalized values
- Hovering shows a tooltip with sample names and values
- Maximum width: 240px (adjusts automatically according to sample count)

Significance criteria (default):

- $\text{FDR} < 0.05$: significant differential expression
- $|\log\text{FC}| > 1$: at least 2-fold expression change (Volcano Plot criterion)
- Upregulated: $\log\text{FC} > 0$ (or > 1 , depending on the plot)
- Downregulated: $\log\text{FC} < 0$ (or < -1 , depending on the plot)

9. PCA (Principal Component Analysis)

This screen provides PCA results for gene expression data through multiple visualizations. It can be used to explore overall similarity/differences among samples, consistency among replicate samples, and genes contributing to each principal component.

⚠ Warning: PCA analysis requires **at least 2 biological replicates for each group**. If this condition is not met, a warning message is displayed along with the current sample structure (example: `"Control (1), Treatment (2)"`), and the analysis screen cannot be accessed.

The PCA screen consists of four sub-tabs:

Sub-tab	Description
PCA Plot	Principal component scatter plot. Used to inspect overall distribution among samples
Sample Correlation	Correlation heatmap across all samples
Replicate Comparison	Group-wise replicate sample quality comparison (4 lower-level tabs)
Gene Loadings	Table of per-gene contributions to each principal component

1. PCA Plot Tab

Visualizes the sample distribution in principal component space using an interactive Plotly.js scatter plot.

Axis selection (above the chart):

- Select the principal component for the X-axis / Y-axis independently using dropdowns (`PC1`, `PC2`, ... `PCn`)
- Default: `X-axis = PC1` , `Y-axis = PC2`
- Variance explained is displayed in each axis label (example: `"PC1 (45.23%)"`)

Chart structure:

- Groups are automatically distinguished using a 12-color palette
- Marker size: `12px` , with outline
- Chart size: fixed at `600 x 600px`

- Plotly toolbar: provides standard tools such as zoom, pan, and screenshot export

Sample Groups panel (below the chart):

- Displays each group name and the sample list in that group
- Includes a circular color indicator for each group

2. Sample Correlation Tab

Visualizes sample correlations as a clustered heatmap. A dendrogram generated by hierarchical clustering with scipy on the server is shown together.

Statistic cards (top, 4 cards):

Card	Description
Total Samples	Total number of samples
Mean Correlation	Mean correlation coefficient
Min Correlation	Minimum correlation coefficient
Max Correlation	Maximum correlation coefficient (excluding diagonal)

Heatmap structure:

- Color scale: **Purple** (low correlation) → **White** (middle) → **Yellow/Gold** (high correlation)
- Row/column dendrogram: based on scipy hierarchical clustering
- Group color bars: the group of each sample is shown by color on the top and left side
- Hovering shows the two sample names and the correlation coefficient
- Chart size: dynamically adjusted according to sample count (minimum 600px, +80px per sample)

Interpretation Guide panel (below the chart):

Correlation coefficient range	Interpretation
≥ 0.95	Very high correlation. Excellent replicate quality
0.85 - 0.95	High correlation. Good replicate quality
0.70 - 0.85	Moderate correlation. Technical variation should be checked
< 0.70	Low correlation. Potential issue or large biological difference

3. Replicate Comparison Tab

Compares consistency among replicate samples within the selected group using multiple methods. A group-selection sidebar is shown on the left, and four lower-level tabs are shown on the right.

Left sidebar (`w-64`):

- `"Select Sample Group"` label
- Group list buttons (click to switch groups)
- The selected group is highlighted in blue

3-1. Fragment Count Sub-tab

Displays fragment (sequencing read) counts for replicate samples in the selected group as a bar chart.

Statistic cards (top, 4 cards):

Card	Description
Total Fragments	Total fragment count of that group
Mean per Sample	Mean fragment count per sample
Min Fragments	Minimum fragment count
Max Fragments	Maximum fragment count

Chart: Plotly bar chart (`700 x 450px`). X-axis: sample name, Y-axis: Fragment Count.

i Reference: If fragment counts vary widely among replicates, a technical issue in library preparation may be suspected.

3-2. Scatter Plot Sub-tab

Compares gene expression among replicate samples using a pairwise scatter plot matrix.

Chart structure:

- `N x N` subplot matrix (`N = number of replicate samples`)
- Diagonal positions: sample name labels
- Off-diagonal positions: expression scatter plots between two samples
- Symmetric axis ranges are applied (same scale across all subplots)
- Chart size: `500 x 500px`
- Up to `1,000` genes are displayed

💡 Interpretation: The more tightly the points cluster along the diagonal line, the more consistent the expression pattern is among replicates. Points deviating from the diagonal indicate differentially expressed genes or technical variation.

3-3. MA Plot Sub-tab

Visualizes differences in expression among replicate samples as an MA plot matrix.

Chart structure:

- $N \times N$ subplot matrix
- X-axis (A): mean $\log_2$ expression of the two replicates
- Y-axis (M): $\log_2$ fold change (difference) between the two replicates
- Diagonal positions: sample name labels
- $M=0$ reference line displayed (gray dotted line)
- Symmetric axis ranges applied
- Chart size: $500 \times 500\text{px}$
- Up to $1,000$ genes are displayed

💡 Interpretation: If points are concentrated around $M=0$, replicate expression is consistent. Points that deviate strongly from $M=0$ are genes with large between-replicate expression differences.

3-4. Correlation Heatmap Sub-tab

Displays the correlation among replicate samples in the selected group as a heatmap.

Statistic cards (top, 3 cards):

Card	Description
Mean Correlation	Mean correlation among replicates
Min Correlation	Minimum correlation coefficient
Max Correlation	Maximum correlation coefficient

Heatmap structure:

- Color scale: **Cyan** (low correlation) → **Light Gray** (middle) → **Magenta** (high correlation)

- Chart size: 500 x 400px
- Hovering shows the two sample names and the correlation coefficient (3 decimal places)

i Reference: The diagonal is always 1.0 (self-correlation). A correlation coefficient ≥ 0.9 indicates acceptable replicate consistency, while < 0.7 suggests a technical issue or biological variation.

4. Gene Loadings Tab

Provides a table of loading values showing how strongly each gene contributes to each principal component (PC). This is used to identify core genes driving a specific principal component.

Control panel (top):

Control	Description	Default / options
Sort by PC	Select the principal component used as the sorting basis	PC1 (choose PC1~PCn from dropdown)
Sort order	Sorting order	Highest absolute value / Lowest absolute value
Genes per page	Number of genes per page	25 / 50 / 100 / 200
Total genes	Displays the total number of genes (right side)	-

Data table:

Column	Description
Gene ID	Gene ID. Left fixed (sticky) column
PC1 ~ PC5	Loading values for each principal component (6 decimal places). Click the header to change sorting
Abs(PCn)	Absolute value of the currently selected sorting PC. Highlighted in blue

- The sorting-basis PC column is emphasized in bold
- Clicking a PC header: if it is the same PC, toggles the sorting order; if it is a different PC, switches to descending sort for that PC
- Server-side sorting and pagination are applied (via API call)

Pagination (bottom):

- Position info in the form "Showing page X of Y (start-end of total)"
- First / Previous / Next / Last buttons

10. Clustering Analysis

This screen performs clustering analysis based on differential expression patterns. Three clustering methods are provided, and they are switched using tabs at the top.

Clustering method	Description
Hierarchical Tree Clustering	DEG-based hierarchical clustering. Dendrogram tree cutting
Mfuzz	Fuzzy c-means-based time-series expression pattern clustering
WGCNA	Weighted gene co-expression network analysis. Module detection

All three methods share the same **three-stage layout** structure:

- **Left:** parameter panel (collapsible/expandable, `w-80 ↔ w-16`)
- **Center:** cluster/module list sidebar (`w-64`)
- **Right:** detailed information for the selected cluster/module (expression pattern chart + gene table)

i Reference: For all clustering methods, the analysis runs automatically with default parameters on first access. If previous results exist, those results are loaded. A full-screen loading overlay is displayed while analysis is running.

1. Hierarchical Tree Clustering

Performs hierarchical clustering based on DEG analysis results, then cuts the tree at a specified height to create subclusters.

Parameter panel:

Parameter	Description	Default
P-value	DEG significance threshold	<code>0.05</code>
Fold Change	Log2 fold change threshold	<code>2</code>
Ptree	Percentile for tree-cut height (<code>1-100</code>). Larger values produce larger clusters, smaller values create more detailed clusters	<code>30</code>
Run Clustering	Button to execute clustering with the configured parameters	<code>-</code>

Cluster sidebar:

- Displays the total number of clusters in the "Clusters (N)" header
- Sorted by cluster ID ascending (Cluster 1, 2, 3...)
- Each item displays the cluster name and gene count
- The selected cluster is highlighted in blue

Cluster detail - expression pattern chart:

- Recharts LineChart (height 500px , responsive width)
- X-axis: sample name, Y-axis: log2(Expression)
- The chart header displays gene count, P-value, Log2 FC, and Ptree parameter information

Chart toggle switches (upper right, 3 switches):

Toggle	Description	Default state
Genes	Show individual gene expression lines (gray semi-transparent, up to 300 genes)	OFF
Mean	Mean expression line (blue solid line, 3px width)	ON
Median	Median expression line (green dashed line, 3px width)	ON

- Tooltip: displays sample name, Mean, and Median values on hover
- Toggle state remains preserved when switching clusters

Cluster detail - gene table:

Column	Description
Checkbox	Gene selection
Gene ID	Gene ID
Gene Symbol	Gene symbol
Expression Pattern	Visualizes expression pattern as a MiniHeatmap
Sample columns	Expression value for each sample

- Search: Gene ID / symbol search (server-side)
- Sorting: by Gene ID
- Rows per page: 100 / 500 / 1,000 / 2,000 / 3,000
- Server-side pagination is applied

Analyze dropdown menu:

Analysis	Description
Heatmap	Expression heatmap of the selected genes
GO Enrichment	Gene Ontology functional analysis
KEGG Pathway	KEGG pathway analysis
Venn Diagram	Venn diagram (moves to the DEG tab)

Download: TSV format. Downloads selected genes or all genes in the cluster.

2. Mfuzz

Time-series expression pattern clustering using the fuzzy c-means algorithm. Each gene is represented by a membership value ($0 \sim 1$) indicating the degree of belonging to a cluster.

Parameter panel - Input Source Type (radio buttons):

Source type	Description	Recommendation
DEG-filtered	Use DEG filtered by P-value / Fold Change ($\sim 2,000$ genes)	-
Variance-filtered	Select top N highly variable genes based on MAD. Best for pattern exploration	Recommended
Full TMM	Use the full TMM-normalized matrix. May introduce noise	Not Recommended

Additional parameters by source type:

Parameter	Source type	Description	Default
P-value	DEG-filtered	Significance threshold	0.05
Fold Change	DEG-filtered	Log2 fold change threshold	2
Top N Genes	Variance-filtered	Number of highly variable genes ($3,000 \sim 10,000$ recommended)	8,000

Clustering parameters:

Parameter	Description	Default
Cluster Count	Number of clusters (c). Range: $2 \sim 20$	16
M value	Fuzzifier parameter. If left blank, it is estimated automatically	auto

Parameter	Description	Default
Min Membership	Minimum membership threshold (0~1). Genes below this value are excluded	0.5
Run Clustering	Button to run clustering	-

Cluster sidebar: same structure as the Tree method. Sorted by cluster ID ascending.

Cluster detail - expression pattern chart: same toggle switch structure (Genes / Mean / Median) as the Tree method.

- The chart header shows Source, P-value/Top genes, Clusters, and Min membership parameter information
- When Genes is toggled ON, only the top 300 genes by membership are displayed

Cluster detail - gene table: same structure as the Tree method, with an additional **Membership** column.

- Membership: degree of belonging to that cluster (0~1 , higher means more representative)

3. WGCNA

Weighted Gene Co-expression Network Analysis. Builds a gene co-expression network and detects modules of genes with similar expression patterns.

Parameter panel - Input Source Type: the same three source types as Mfuzz (DEG-filtered / Variance-filtered / Full TMM).

- Default Top N Genes for Variance-filtered: 5,000 (smaller than Mfuzz's 8,000 , recommended range 3,000~5,000)

WGCNA-specific parameters:

Parameter	Description	Default
Soft Thresholding Power	Controls network connection weighting. Auto-detect or manual input (1~30)	Auto-detect
Run WGCNA Analysis	Button to run the analysis	-

Advanced Parameters (collapsible/expandable):

Parameter	Description	Default
Min Module Size	Minimum number of genes per module (10~200)	30
Deep Split	Cluster splitting sensitivity (0~4). Larger values create finer modules	2

Parameter	Description	Default
Merge Cut Height	Similar-module merge threshold (<code>0~1</code>). Smaller values merge more aggressively	<code>0.25</code>

Module sidebar (WGCNA-specific):

- `"Modules (N)"` header
- Sorted by gene count descending (largest module first)
- Each module name uses the standard WGCNA color name (`turquoise` , `blue` , `brown` , etc.)
- Displays a module color indicator (`8x8` color box)
- **Gene search:** enter a Gene ID in the search box to filter only modules containing that gene
- Search result state is displayed (`"Showing modules with: [search term]"`)
- Click the `X` button to clear the search

Module detail - Module Eigengene Pattern chart:

- Recharts `LineChart` (height `400px` , responsive width)
- Module Eigengene line: thick solid line (`3px`) using the representative module color
- The chart header displays the module color indicator, gene count, and parameter information
- Show Genes toggle: when ON, the top 100 genes by kME are displayed as individual lines (gray semi-transparent)

i Reference: Instead of Mean/Median statistics used in Tree/Mfuzz, WGCNA uses the Module Eigengene (the first principal component of the module) as the central trend.

Module detail - gene table: same base structure as the Tree method, with an additional WGCNA-specific column.

Additional column	Description
Module Membership (kME)	Correlation with the module eigengene. The higher the value, the more core the gene is within the module

4. Common Interaction Pattern (All Clustering Methods)

Collapse/expand of parameter panel:

- Expanded state: `w-80 (320px)` . Parameter inputs and Run button are displayed
- Collapsed state: `w-16 (64px)` . Only vertical `"Params"` text and the arrow icon are shown

- Collapse/expand is toggled using the thin vertical button on the right (**w-8**)
- Clicking in the collapsed state expands the panel

Gene selection and analysis:

- Select genes using checkboxes in the gene table
- Selection state is reset when switching cluster/module
- Analyze menu: **Heatmap** , **GO Enrichment** , **KEGG Pathway** , **Venn Diagram**
- Download: TSV format

Behavior when parameters change:

- Parameter change → existing results are retrieved automatically (if cached results exist, they are displayed immediately)
- Click Run button → executes a new analysis (full-screen loading overlay)
- Analysis may take several minutes depending on dataset size

11. Heatmap

This screen visualizes gene expression as a color matrix. User-defined gene sets can be managed, and heatmaps can be generated using multiple normalization methods and clustering options.

Screen Layout

Area	Position	Description
Interesting Gene Sets sidebar	Left (280px , horizontally resizable)	Gene set tree management
Heatmap controls	Upper right	Normalization / clustering options
Heatmap visualization	Right main area	Plotly.js-based heatmap

Interesting Gene Sets Sidebar

This sidebar manages gene sets in a folder/file tree structure.

Header:

- "Interesting Gene Sets" title
- + **button**: opens the new gene set creation modal

Search:

- Typing text into the search box performs real-time filtering by file name
- Parent folders of matching items expand automatically
- Matching parts are highlighted in yellow
- Use the X button to clear the search

Tree structure:

- **Folder** (yellow folder icon): click to expand/collapse. Top-level folders expand automatically on mount
- **File** (document icon): gene set file (.txt)

File selection:

- Single click: selects only that file → displays the genes from that file in the heatmap

- **Ctrl+Click**: toggles multi-selection. Genes from multiple files are merged and displayed in the heatmap (duplicates removed)
- Selected files are highlighted with a blue background
- A **"Ctrl+Click to select multiple"** helper text is shown at the bottom of the sidebar

Mouse-hover actions:

Target	Button shown on hover	Action
File	Pencil icon (Edit)	Open the gene set edit modal
File	Trash icon (Delete)	Delete after confirmation
Folder	Trash icon (Delete)	Delete the folder after confirmation

 **Warning:** Deleting a folder also deletes all files inside it.

Gene Set Editor Modal

This modal is used to create or edit gene sets. It is opened by the + button or a file edit icon.

Top section of the modal - select save location:

- Select the save location from the folder tree (click to select, background highlight)
- **New Folder** button: create a new folder under the selected folder
 - Inline input field appears → enter folder name → **Enter** to confirm, **Escape** to cancel
- The last selected folder is saved in localStorage and selected automatically next time

Create Mode

Two input tabs are provided:

Tab 1 - Text Input (manual input):

Field	Description
File Name	Gene set file name (.txt is added automatically)
Gene IDs	Enter one Gene ID per line in the text area

- **Create** button: creates a new file using the entered genes

Tab 2 - Upload Files (file upload):

Button	Description
Choose Files	Select <code>.txt</code> files individually (multiple selection allowed)
Choose Folder	Select a whole folder → sub- <code>.txt</code> files are detected automatically

- The selected file list is displayed as a tree preview (file name + number of genes)
- **Upload** button: uploads the selected files to the specified folder

Edit Mode

- The gene list of the existing file is loaded into the text area
- The file name is shown as read-only
- Add/remove Gene IDs and save with the **Save** button

Folder Management

Folders can also be deleted inside the modal:

- A trash icon appears when hovering over a folder
- Clicking it deletes that folder and its child files after confirmation

Heatmap Controls

When a gene set is selected, normalization and clustering options are shown above the heatmap.

Normalization (dropdown, 3 methods):

Method	Description
Z-score	Scales each gene to mean=0, standard deviation=1. Suitable for comparing relative patterns among genes
Log2 Centered	Applies log2 transformation and subtracts the gene-wise mean
Log2FC vs Reference	Calculates log2 fold change relative to the selected reference sample

Reference Sample (dropdown):

- Displayed only when `Log2FC vs Reference` is selected
- Select the sample to use as the reference

Clustering (dropdown, 3 methods):

Method	Description
Ward	Ward minimum-variance criterion. Tends to create similarly sized clusters
Average	Average linkage (UPGMA). Balanced clustering
Complete	Complete linkage. Tends to create compact clusters

- The selected clustering method rearranges the gene (row) order in the heatmap

Heatmap Visualization

Interactive heatmap based on Plotly.js.

Title:

- Displays the selected gene set file name (lists all names when multiple files are selected)
- Shows the number of genes and the applied normalization method

Color scale:

- Applies the colorscale configured in Settings
- In Z-score mode: `zmid=0`, and the range is clipped automatically using the absolute value of the 99th percentile of the data (minimum 2, maximum 5)
- A color bar is shown on the right side of the heatmap

Dynamic size adjustment:

- Height: adjusted automatically by gene count (minimum `300px` ~ maximum `1,200px`, `2~15px` per gene)
- Width: responsive to the container

Interactions:

Action	Description
Drag	Zoom into a selected area
Double click	Reset zoom (restore full view)
Hover	Display gene name, sample name, and expression value tooltip
Click sample (Log2FC mode)	Change the clicked sample to the new reference sample

When there are many genes - use zoom:

If the number of genes is large, Y-axis labels may overlap, making gene names hard to read. In this case, the zoom function can be used to inspect gene names clearly within a specific range.

- **Drag the mouse** on the heatmap to zoom into the selected area
- **Double-click** to return to the full view
- As fewer genes remain visible after zooming, the Y-axis font size increases automatically

Number of genes displayed after zoom	Y-axis font size
≤10	14px
≤20	12px
≤50	10px
≤100	8px
>100	6px

Image download:

- Click the camera icon in the Plotly toolbar → download a high-resolution PNG (10x scale)
- Helper text below the heatmap: "Drag to zoom, double-click to reset | Use camera icon in toolbar to download high-resolution PNG"

Caching:

- Results for the same combination of gene set + normalization + clustering are cached in memory
- If an option is changed and then reverted, the result is displayed immediately without another API call

Heatmap Analysis Modal

This is a modal-form heatmap opened from the **Analyze** → **Heatmap** menu in the DEG tab or Clustering tab.

Item	Description
Title	"Expression Heatmap - {comparison group name}"
Width	Fixed at 900px
Controls	Normalization, Reference Sample, Clustering — same as the main Heatmap screen

Item	Description
Interactions	Zoom, hover, reference sample change, automatic font adjustment — same as the main Heatmap screen
Close	Bottom Close button or click outside the modal

 **Reference:** This modal generates the heatmap from the gene list selected in that analysis, not from Interesting Gene Sets.

12. Venn Diagram

This screen visually compares the intersections and differences among up to three gene sets. By clicking each region of the Venn diagram, the corresponding expression matrix can be viewed, and GO/KEGG/Heatmap analyses can be performed.

Screen Layout

Area	Position	Description
Gene Sets input panel	Left (224px , collapsible)	Gene set input/management
Venn Diagram	Center (500x500px)	Highcharts-based Venn diagram
Gene result table	Right (remaining area)	Expression matrix for the selected region

Gene Sets Input Panel

A collapsible left panel used to enter and manage gene sets.

Panel elements (per set):

Element	Description
Set name	Editable through text input (defaults: A, B, C)
clear button	Delete genes only (keep the set itself)
delete button	Delete the entire set
Text area	Enter Gene IDs (height 192px). Line breaks, commas, and semicolons are supported
Gene count display	Displays the number of entered genes in real time

Buttons at the bottom of the panel:

- **+ Add Set:** add a new set (up to 3 sets)
- **Reset All Sets:** reset all sets (red button)

Collapse/expand:

- Collapse is toggled using the button on the far-right edge

- Collapsed state: only vertical "Gene Sets" text and an expand icon are shown

Reference: Gene set data is managed in the parent component (`WorkbenchDetail`), so the entered content is preserved when switching to other tabs and returning. Genes can also be passed from other tabs through the **Analyze** → **Venn Diagram** menu in the DEG/Clustering tabs.

Venn Diagram

Interactive diagram based on the Highcharts Venn module.

Display conditions:

- At least 2 gene sets are required
- Each set must contain at least 1 gene
- If the conditions are not satisfied, a guidance message is shown

2-set diagram (`A`, `B`):

Region	Description
A Only	Genes existing only in A
B Only	Genes existing only in B
A ∩ B	Intersection of A and B

3-set diagram (`A`, `B`, `C`):

Region	Description
A Only	Genes existing only in A
B Only	Genes existing only in B
C Only	Genes existing only in C
A ∩ B	Intersection of A and B (excluding C)
B ∩ C	Intersection of B and C (excluding A)
A ∩ C	Intersection of A and C (excluding B)
A ∩ B ∩ C	Intersection shared by all three sets

Region click interaction:

1. Click a circle or overlapping area in the diagram
 2. A context menu appears at the mouse position:
 - ○ **Exclusive:** genes belonging only to that region (genes not present in other sets)
 - ● **Total:** all genes included in that set (including genes overlapping with other sets)
 - The number of genes is shown next to each option
 3. Select an option → the expression data for those genes is loaded into the table on the right
- **Ctrl+Click:** adds new genes to the existing selection (accumulative selection)

Hover tooltip: shows the number of Exclusive/Total genes for that region

Diagram download: PNG, JPEG, PDF, or SVG can be downloaded from the top of the diagram or from the Highcharts export menu

Gene Result Table

Expression matrix table displayed after a Venn diagram region is selected.

Toolbar above the table:

Element	Description
Matrix Type	Buttons for Raw / TPM / TMM. Selecting one requeries the corresponding matrix from the server
Download	Download expression data of selected genes as CSV (green button)
Analyze	Dropdown menu (Heatmap / GO Enrichment / KEGG Pathway)

Matrix Type descriptions:

Type	Description
Raw	Raw read count
TPM	Transcripts Per Million. Suitable for inter-sample comparison
TMM	Trimmed Mean of M-values. Normalized expression value

Table columns:

Column	Description	Sticky
Checkbox	Gene selection (use the header checkbox to select the entire current page)	Fixed on the left

Column	Description	Sticky
#	Row number	Fixed on the left
Gene ID	Gene ID (Gene Description tooltip on hover)	Fixed on the left
Gene Symbol	Gene symbol (italic when different from Gene ID)	Fixed on the left
Expression Pattern	Visualized as a MiniHeatmap (16px per sample, up to 240px)	-
Sample columns	Expression value for each sample	-

- The first 4 columns on the left (Checkbox, #, Gene ID, Gene Symbol) remain fixed during horizontal scrolling
- The header remains fixed during vertical scrolling

Pagination:

- Displayed at both the top and bottom of the table
- Rows per page: 100 / 500 / 1,000 / 2,000 / 5,000
- Current range display (example: "1-500 of 2,543")
- Page number buttons (long lists are collapsed with ellipsis) + Previous/Next buttons

Empty state:

- No region selected: displays the message "Click on the Venn diagram to view genes"

Analyze Dropdown Menu

Analysis functions available after genes are selected using checkboxes.

Analysis	Description
Heatmap	Display a heatmap modal of the selected genes
GO Enrichment	Gene Ontology functional analysis (BP, MF, CC categories)
KEGG Pathway	KEGG pathway analysis

 **Reference:** Genes must first be selected in the table before the analysis can be run.